WAVE PACKET DENSITY IN LOCAL SMOOTHING: REFINEMENTS AND LIMITATIONS

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Abstract. We study wave packet density estimates in the local smoothing problem for the wave equation. Building on the framework of Gan, He, Li, and Wu, we refine the broad-incidence argument to obtain an improved local smoothing range for wave equations on compact Riemannian manifolds in even dimensions. We then investigate the limitations of wave packet density methods, showing that genuine geometric obstructions already arise in the Euclidean setting.

1. Introduction

Local smoothing estimates measure the gain in regularity that arises when a solution to a dispersive equation is averaged in time. For the wave equation, let

$$u(x,t) = e^{it\sqrt{-\Delta}} f(x)$$

denote the half-wave evolution on $\mathbb{R}^n$. At each fixed time, the classical estimates of Peral and Seeger–Sogge–Stein show that the half-wave propagator loses

$$s_p = (n-1)\left|\frac{1}{2} - \frac{1}{p}\right|$$

derivatives in L^p [8, 9]. Sogge observed that averaging over a compact time interval produces additional regularity and formulated the local smoothing conjecture [10]. In the range $p > 2$, the conjecture predicts that, for every $\varepsilon > 0$,

$$\|e^{it\sqrt{-\Delta}} f\|_{L^p(\mathbb{R}^n \times [1,2])} \lesssim_\varepsilon \|f\|_{L^p_{s_p - \frac{1}{p} + \varepsilon}(\mathbb{R}^n)}$$

whenever

$$p \geqslant \frac{2n}{n-1}.$$

Thus, compared with the fixed-time estimate, one expects an almost $1/p$ derivative gain.

The local smoothing problem is closely connected with restriction theory, Kakeya-type phenomena, and oscillatory integral estimates. Wolff initiated a systematic approach to the problem through decoupling inequalities for the cone [13]. A major breakthrough was the sharp ℓ^2 decoupling theorem of Bourgain and Demeter [2], which implies the local smoothing conjecture in the range

$$p > 2 + \frac{4}{n-1}.$$

The corresponding variable-coefficient estimates for Fourier integral operators satisfying the cinematic curvature condition were established by Beltran, Hickman, and Sogge [1]. In two spatial dimensions, Guth, Wang, and Zhang proved the full Euclidean local smoothing conjecture by establishing a sharp square function

estimate for the cone [6]; the corresponding result for Fourier integral operators and wave equations on compact Riemannian surfaces was subsequently obtained by Gao, Liu, Miao, and Xi [5].

In dimensions $n \geqslant 3$, substantial further progress was recently obtained by Gan, He, Li, and Wu [4]. Their work introduces a wave packet density adapted to Lorentz rescaling and combines refined decoupling, broad estimates, and two-ends reductions. This provides a new framework for the local smoothing problem that goes beyond the decoupling exponent. For wave equations on general compact Riemannian manifolds, Minicozzi and Sogge constructed examples showing that the local smoothing estimate cannot hold below

$$p_n^+ := \begin{cases} 2 + \dfrac{8}{3n-3}, & n \text{ odd}, \\ 2 + \dfrac{8}{3n-2}, & n \text{ even}, \end{cases} \tag{1.1}$$

see [7]. Gan–He–Li–Wu proved the conjectured sharp range $p > p_n^+$ in all odd dimensions. In even dimensions, they obtained the estimate for

$$p > 2 + \frac{8}{3n-2-\omega(n)}, \qquad \omega(n) = \frac{4(7n-2)}{3n^2+5n+14}, \tag{1.2}$$

which is within $O(n^{-3})$ of the conjectured critical exponent.

The first purpose of this paper is to further refine the framework of [4]. Our argument separates the k-broad structure governing the L^p estimate from the 2-broad structure used in the hairbrush argument, and combines this refinement with an additional multilinear geometric estimate and a standard L^2 estimate. As a consequence, for wave equations on compact Riemannian manifolds in every even dimension $n \geqslant 4$, we improve the exponent of Gan–He–Li–Wu from (1.2) to

$$p > p'(n) := 2 + \frac{8}{3n-2-\omega'(n)}, \qquad \omega'(n) := \frac{12(n+2)}{2n^2+7n+14}.$$

In particular, $p'(4) = 482/167 = 2.8862\ldots$.

It is also natural to ask whether substantially stronger local smoothing estimates can be obtained in the Euclidean setting by exploiting additional geometric information specific to flat space. The second purpose of this paper is to investigate this question through an explicit hyperbolic-type configuration of wave packets. We first show that this configuration nearly saturates the original wave packet density estimate and yields sharp obstructions corresponding to the conjectured critical exponents for local smoothing on manifolds. We then examine an anisotropic multiscale version of wave packet density. Although the multiscale formulation captures finer geometric information and weakens the original obstruction, the same configuration still produces a nontrivial barrier. These results illustrate structural limitations of wave packet density methods in the Euclidean local smoothing problem.

• **Structure of the article.** In Section 2, we recall some key concepts, quantities, and estimates from [4]. In Section 3, we obtain a refined local smoothing range in high even dimensions. In Section 4, we construct a nearly extremal example in the Euclidean setting that reveals a geometric obstruction already present in flat space, illustrating a limitation of the current form of the wave packet density approach to the local smoothing problem.

• **Notation and Parameters.** Throughout this paper, we adopt the notation, definitions, quantities, and assumptions introduced in [4]. We use $\lessapprox$ for an arbitrarily small power loss in R. In the proof of the refinement, these losses are chosen sufficiently small to be included in $R^{O(\varepsilon^2)}$. The parameters satisfy $0 < \delta \ll \varepsilon^{500} \ll 1$. As usual, rapidly decreasing wave packet tails are suppressed; all local estimates may be applied with fixed smooth cutoffs and slightly enlarged sets.

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2. Preliminaries

In this section, we recall some key concepts from [4] and introduce a multilinear geometric estimate.

2.1. **Wave Packet Density.**

The idea underlying wave packet density can be traced back to the wave envelope estimates of Guth, Wang, and Zhang [6], where the geometric concentration of wave packets is captured through localized square function estimates at multiple scales. Building on this perspective, Gan, He, Li, and Wu [4] further developed and formalized the notion of wave packet density as an L^2-based quantity compatible with Lorentz rescaling, incorporating it into a systematic framework for higher-dimensional local smoothing estimates.

Following [4], we use wave packet density as an L^2-based substitute for $\|f\|_\infty$. Rather than measuring the pointwise size of f, this quantity measures the concentration of the L^2 mass of its wave packets. Its definition in terms of nonnegative sums of squared L^2 norms makes it compatible with almost orthogonality and restriction to subcollections of wave packets, and hence convenient for induction on scales.

For an s-cap $\tau \in \Theta_s$, where

$$R^{-1/2} \leqslant s \leqslant 1,$$

let $V_{\tau,R} := (Rs^2)\tau^*$, where τ^* denotes the dual box associated with τ. Thus, $V_{\tau,R}$ is a box centered at the origin with dimensions $Rs^2 \times Rs \times \cdots \times Rs$. We write $V \parallel V_{\tau,R}$ when V is a translate of $V_{\tau,R}$. As in [4], $\mathbb{T}_\theta(z_0)$ denotes the collection of wave packet planks associated with θ at scale R in $B_R^{n+1}(z_0)$, and $T^\flat$ denotes the base box associated with T.

We recall the following definition from [4, Definition 4.1].

Definition 2.1 (Wave packet density)**.** *Let $g \in \mathcal{S}(\mathbb{R}^n)$ satisfy $\operatorname{supp}\widehat{g} \subset B_1^n \backslash B_{1/2}^n$, and let*

$$B_R^{n+1}(z_0) = B_R^{n+1}(x_0, t_0) \subset B_\lambda^{n+1}.$$

The wave packet density of g in $B_R^{n+1}(z_0)$ is defined by

$$\begin{aligned} &\mathcal{W}(g, B_R^{n+1}(z_0)) \\ &:= \sup_{R^{-1/2} \leqslant s \leqslant 1} \sup_{\tau \in \Theta_s} \sup_{V \parallel V_{\tau,R}} \left(\frac{1}{|V|} \sum_{\substack{\theta \in \Theta_{R^{-1/2}} \\ \theta \subset \tau}} \sum_{\substack{T \in \mathbb{T}_\theta(z_0) \\ T^\flat \subset V}} \|g_T\|_{L^2(\mathbb{R}^n)}^2 \right)^{1/2}. \end{aligned} \tag{2.1}$$

Let $\mathcal{F}$ be a type-**1** Fourier integral operator, with the phase and amplitude normalized as in [4, Sections 3.1–3.2], and let $\mathcal{F}^\lambda$ denote its rescaled version. The main result of [4] is the following.

Theorem 2.2. *For every $\varepsilon > 0$ and*

$$p \geqslant p(n) := \begin{cases} p_n^+, & n \text{ odd}, \\ 2 + \dfrac{8}{3n - 2 - \omega(n)}, & n \text{ even}, \end{cases} \qquad \omega(n) := \frac{4(7n-2)}{3n^2+5n+14},$$

one has

$$\|\mathcal{F}^\lambda f\|_{L^p(B_R^{n+1})} \lesssim_\varepsilon R^{(n-1)\left(\frac{1}{2}-\frac{1}{p}\right)+\varepsilon} \|f\|_2^{\frac{2}{p}} \mathcal{W}(f, B_R^{n+1})^{1-\frac{2}{p}}, \tag{2.2}$$

for every $R \in [1, \lambda^{1-\varepsilon}]$ and all $f \in \mathcal{S}(\mathbb{R}^n)$ with

$$\begin{aligned} \operatorname{supp} \widehat{f} \subset \mathbb{A}^n(1) := \big\{ (\xi_1, \dots, \xi_n) \in B_1^n \backslash B_{1/2}^n : \xi_j > 0 \text{ for } 1 \leqslant j \leqslant n-1, \\ |(\xi_1, \dots, \xi_{n-1})| \leqslant \xi_n \big\}. \end{aligned} \tag{2.3}$$

For the higher-dimensional refinement, we impose one additional condition on the phase. For each fixed z, suppose that there is a symmetric, nondegenerate $(n+1) \times (n+1)$ matrix $\mathbf{Q}_z$, independent of ξ, such that

$$\big(\partial_z \phi(z;\xi)\big)^T \mathbf{Q}_z \, \partial_z \phi(z;\xi) = 0. \tag{2.4}$$

We call this the *quadratic-cone condition*. On normalized coordinate patches, the matrices may be normalized so that $\|\mathbf{Q}_z\| + \|\mathbf{Q}_z^{-1}\| \lesssim 1$; see Lemma 2.7 below. This condition holds for local parametrices of Riemannian half-wave operators, by the eikonal equation. It is not assumed for the general FIO theorem quoted above.

Theorem 2.3. *Let $n \geqslant 4$ be even, and let $\mathcal{F}$ be a type-**1** FIO as above satisfying* (2.4)*. For every $\varepsilon > 0$, the estimate* (2.2) *holds for all $p \geqslant p'(n)$, all $R \in [1, \lambda^{1-\varepsilon}]$, and all $f \in \mathcal{S}(\mathbb{R}^n)$ satisfying* (2.3)*.*

In particular, the usual local smoothing estimate for Riemannian wave equations holds for $p > p'(n)$ in every even dimension $n \geqslant 4$. The passage from the mixed estimate to local smoothing is the standard reduction in [4, Sections 3.1 and 4].

The mixed wave packet density estimate is stronger than the corresponding local smoothing estimate in the Euclidean setting. The density records the L^2 mass of packets whose base boxes lie in a matching test box; it does not directly count all spacetime incidences. For particular configurations, this distinction makes the mixed estimate strictly more restrictive than the corresponding local smoothing estimate. We construct a hyperbolic-type configuration showing that the conjectured critical exponent for local smoothing on manifolds provides a necessary lower bound for the original wave packet density estimate even in the Euclidean setting.

Theorem 2.4. *Let $n \geqslant 3$ and $R \gg 1$. Then there exist functions f_R with*

$$\operatorname{supp} \widehat{f_R} \subset B_1^n \backslash B_{1/2}^n$$

such that for all $\frac{2n}{n-1} < p < p_n^+$

$$\left\| e^{it\sqrt{-\Delta}} f_R \right\|_{L^p(B_R^{n+1})} \gtrsim R^{(n-1)\left(\frac{1}{2}-\frac{1}{p}\right)+c_{n,p}} \|f_R\|_2^{\frac{2}{p}} \mathcal{W}(f_R, B_R^{n+1})^{1-\frac{2}{p}}$$

for some constant $c_{n,p} > 0$.

We next consider an anisotropic multiscale version of wave packet density suggested in [4, Remark 4.9]. Let

$$R^{-1/2} \leqslant s_1, \dots, s_{n-1} \leqslant 1$$

be dyadic numbers, and set $s_{\max} := \max_{1\leqslant i\leqslant n-1} s_i$. Let $\tau \subset B^n_1 \backslash B^n_{1/2}$ be an anisotropic cap of dimensions $1 \times s_1 \times \cdots \times s_{n-1}$. We define $V_{\tau,R}$ to be the dual box centered at the origin with dimensions $Rs^2_{\max} \times Rs_1 \times \cdots \times Rs_{n-1}$ with edges oriented according to the corresponding directions of τ.

Suppose that $g \in \mathcal{S}(\mathbb{R}^n)$ satisfies $\operatorname{supp} \widehat{g} \subset B^n_1 \backslash B^n_{1/2}$. The *multiscale wave packet density* of g in $B^{n+1}_R(z_0)$ is defined by

$$\widetilde{\mathcal{W}}(g, B^{n+1}_R(z_0)) := \sup_{\substack{R^{-1/2}\leqslant s_1,\dots,s_{n-1}\leqslant 1 \\ s_1,\dots,s_{n-1} \text{ dyadic}}} \sup_{\tau} \sup_{\substack{V \| V_{\tau,R} \\ V \subset B^n_R}} \left(\frac{1}{|V|} \sum_{\substack{\theta \in \Theta_{R^{-1/2}} \\ \theta \subset \tau}} \sum_{\substack{T \in \mathbb{T}_\theta(z_0) \\ T^\flat \subset V}} \|g_T\|_2^2 \right)^{1/2},$$

where the second supremum is taken over all anisotropic $(s_1, \dots, s_{n-1})$-caps τ.

Although this multiscale density captures finer geometric information, the configuration constructed above still yields a nontrivial obstruction.

Theorem 2.5. *Let $R \gg 1$, and f_R be the function constructed in Theorem 2.4 when $n = 3$. Then, for every*

$$3 < p < \frac{22}{7},$$

we have

$$\left\| e^{it\sqrt{-\Delta}} f_R \right\|_{L^p(B^4_R)} \gtrsim R^{2\left(\frac{1}{2}-\frac{1}{p}\right)+c_p} \|f_R\|_2^{\frac{2}{p}} \widetilde{\mathcal{W}}(f_R, B^4_R)^{1-\frac{2}{p}}$$

for some constant $c_p > 0$.

This example shows that the multiscale formulation does not completely eliminate the obstruction arising from the original configuration.

2.2. **Broad Reduction.** In this section, we recall the broad–narrow reduction [3], also known as the Bourgain–Guth method.

Let $K = R^{\varepsilon^{50}}$. We say that a K^2-ball is k-narrow if the wave packets intersecting it are concentrated near a $(k-1)$-dimensional subspace. Otherwise, we say that it is k-broad. Let X be the union of all k-broad K^2-balls $B^{n+1}_{K^2} \subset B^{n+1}_R$.

We now give the formal definition of the broad norm. Fix a sufficiently large integer A and $1 \leqslant k \leqslant n+1$. For a K^2-ball $B^{n+1}_{K^2} \subset B^{n+1}_\lambda$ centered at z_0, define

$$\mu_{\mathcal{F}^\lambda f}(B^{n+1}_{K^2}) := \min_{V_1,\dots,V_A} \max_{\tau \notin V_i} \int_{B^{n+1}_{K^2}} \left| \mathcal{F}^\lambda f_\tau \right|^p.$$

Here, $V_1, \dots, V_A$ range over all $(k-1)$-dimensional subspaces, and $\tau \notin V_i$ refers to those $\tau \in \Theta_{K^{-1}}$ satisfying

$$\angle\big(G^\lambda(z_0;\tau), V_i\big) > K^{-2}$$

for all $1 \leqslant i \leqslant A$, where the angle of a set of directions with a subspace is the infimum of the individual angles. The maximum over an empty family is defined

to be zero. If $U \subset B^{n+1}_\lambda$ is a disjoint union of K^2-balls, we define the broad norm by

$$\|\mathcal{F}^\lambda f\|^p_{BL^p_{k,A}(U)} := \sum_{B^{n+1}_{K^2} \subset U} \mu_{\mathcal{F}^\lambda f}(B^{n+1}_{K^2}).$$

By the standard rescaling argument, if, uniformly over the relevant normalized phase class, for every $f \in \mathcal{S}(\mathbb{R}^n)$ satisfying (2.3), one has

$$\|\mathcal{F}^\lambda f\|_{BL^p_{k,A}(X)} \lesssim_\varepsilon R^{(n-1)\left(\frac{1}{2}-\frac{1}{p}\right)+\varepsilon} \|f\|_2^{\frac{2}{p}} \mathcal{W}(f, B^{n+1}_R)^{1-\frac{2}{p}},$$

then the linear estimate (2.2) follows provided that

$$p \geqslant 2 + \frac{4}{2n - \max\{3, k\} + 1}. \tag{2.5}$$

2.3. **Stability of the quadratic-cone condition.** The broad–narrow reduction applies the induction hypothesis to Lorentz-rescaled phases. We therefore verify that the additional condition (2.4) is preserved under these reductions, with uniform quantitative bounds after normalization.

Lemma 2.6 (Stability under rescaling)**.** *Suppose that ϕ satisfies* (2.4)*. Then the phases arising from the local changes of variables and Lorentz rescaling in* [4, Sections 3.1–3.2 and 5.1] *also satisfy the quadratic-cone condition. On the resulting fixed-size normalized positive-curvature patches, the defining matrices can be chosen so that*

$$\|\widetilde{\mathbf{Q}}_w\| + \|\widetilde{\mathbf{Q}}_w^{-1}\| \lesssim 1,$$

with constants depending only on the normalized phase bounds, not on the rescaling parameters.

Proof. Write $v(z;\xi) = \partial_z \phi(z;\xi)$.

Preservation of the quadratic relation. Consider a transformed phase of the form

$$\widetilde{\phi}(w;\eta) = c\,\phi(\Psi(w); \Xi(\eta)) + \psi(\eta), \tag{2.6}$$

where $c \neq 0$, Ψ is a spacetime diffeomorphism, Ξ is a homogeneous frequency diffeomorphism independent of w, and ψ depends only on η. Setting $J(w) = D\Psi(w)$, the chain rule gives

$$\widetilde{v}(w;\eta) := \partial_w \widetilde{\phi}(w;\eta) = cJ(w)^T v(\Psi(w); \Xi(\eta)).$$

Define

$$\widetilde{\mathbf{Q}}_w = J(w)^{-1} \mathbf{Q}_{\Psi(w)} J(w)^{-T}. \tag{2.7}$$

Then

$$\widetilde{v}(w;\eta)^T \widetilde{\mathbf{Q}}_w \widetilde{v}(w;\eta) = c^2 v(\Psi(w); \Xi(\eta))^T \mathbf{Q}_{\Psi(w)} v(\Psi(w); \Xi(\eta)) = 0.$$

Moreover, $\widetilde{\mathbf{Q}}_w$ is symmetric, independent of η, and nondegenerate, since

$$\det \widetilde{\mathbf{Q}}_w = \frac{\det \mathbf{Q}_{\Psi(w)}}{(\det J(w))^2} \neq 0.$$

The Lorentz rescaling used in [4, Section 5.1] is of the form (2.6). To see this explicitly, write the center of the frequency cap as $(\omega, 1)$ and use the phase coordinates from [1, Section 2.5]. Let

$$\Psi_\omega(y,t) = (X_\omega(y,t), t)$$

be determined locally by

$$\partial_{\xi'}\phi(X_\omega(y,t),t;\omega,1)=y',\qquad \phi(X_\omega(y,t),t;\omega,1)=y_n.$$

Before the final bounded linear normalization, the rescaled phase has the exact expression

$$\widetilde{\phi}(y,t;\eta)=K^2\phi\left(\Psi_\omega(K^{-1}y',K^{-2}y_n,t);K^{-1}\eta'+\eta_n\omega,\eta_n\right);$$

see [1, equation (2.14)]. Thus (2.7) applies. The subsequent linear changes of variables and localization preserve the same condition. Likewise, subtracting $\phi(0;\xi)$ does not change $\partial_z\phi$, and the dilation

$$\phi^\lambda(z;\xi)=\lambda\phi(z/\lambda;\xi)$$

satisfies the quadratic-cone condition with defining matrix $\mathbf{Q}_{z/\lambda}$.

Uniform nondegeneracy after normalization. The preceding argument establishes exact preservation, but an arbitrary non-isotropic change of variables need not preserve the condition number of a particular defining matrix. We therefore normalize the defining form on each normalized phase patch.

At a fixed spacetime point, the usual positive-curvature normalization allows the frozen cone to be parametrized, after a uniformly bounded linear change of covector coordinates, by

$$r(\eta,1,h(\eta)),\qquad r>0,$$

where, on a ball $|\eta|<a_0$ of fixed radius,

$$h(0)=0,\qquad \nabla h(0)=0,\qquad c_0I\leqslant\nabla^2h(\eta)\leqslant C_0I.$$

The constants a_0,c_0,C_0 depend only on the normalized phase bounds; compare [1, Lemma 2.5] and the normalization in [4, Lemma 5.7].

Let $\mathbf{M}$ be a nondegenerate symmetric matrix defining this cone. At $\eta=0$, the cone has radial vector e_n and tangent space $\operatorname{span}\{e_1,\dots,e_n\}$. Consequently, $\mathbf{M}e_n$ is a nonzero multiple of e_{n+1}. Multiplying $\mathbf{M}$ by a nonzero scalar, we may assume that $\mathbf{M}e_n=-e_{n+1}$. Differentiating the quadratic relation twice at $\eta=0$ then gives

$$\mathbf{M}=\begin{pmatrix}H&0&b\\0&0&-1\\b^T&-1&d\end{pmatrix},\qquad H=\nabla^2h(0)\asymp I.$$

The equation of the cone becomes

$$\eta^TH\eta-2h(\eta)+2h(\eta)b\cdot\eta+d\,h(\eta)^2=0. \tag{2.8}$$

Fix $0<a<a_0$, independently of the rescaling parameters, and set

$$h_i^\pm:=h(\pm ae_i)\asymp a^2,\qquad 1\leqslant i\leqslant n-1.$$

Evaluating (2.8) at $\pm ae_i$ and dividing by $2h_i^\pm$, we obtain

$$\pm ab_i+\frac{d}{2}h_i^\pm=1-\frac{a^2H_{ii}}{2h_i^\pm}.$$

The right-hand sides are uniformly bounded. Adding the two identities first gives a uniform bound for d, since $h_i^++h_i^-\asymp a^2$, and substitution then gives uniform bounds for each b_i. Hence $\|\mathbf{M}\|\lesssim 1$. Since

$$\det\mathbf{M}=-\det H$$

and $H\asymp I$, we also have $\|\mathbf{M}^{-1}\|\lesssim 1$. Undoing the uniformly bounded covector change preserves these bounds.

Finally, the Lorentz rescaling and subsequent localization return the phase to the same quantitative class, as in [1, Lemma 2.3 and Remark 2.4]. Applying the preceding normalization to each resulting phase gives uniform bounds for its defining quadratic form. These bounds depend only on the fixed normalized phase constants and do not accumulate with the number of rescaling steps. □

Thus the broad–narrow induction may be carried out within the normalized subclass satisfying (2.4). The transformed quadratic form need not coincide with the original one, nor retain the block-diagonal form associated with the original Riemannian metric.

2.4. **A multilinear geometric estimate.** We introduce a multilinear geometric estimate that will be used to improve the incidence estimate. This estimate generalizes Lemma 5.8 of [4].

Lemma 2.7 (Gauss directions and quadratic cones)**.** *Assume* (2.4). *Write* $v(z;\xi) = \partial_z \phi(z;\xi)$. *Then*

$$G(z;\xi) = \pm \frac{\mathbf{Q}_z v(z;\xi)}{|\mathbf{Q}_z v(z;\xi)|}. \tag{2.9}$$

Consequently, at each fixed z, *for* $1 \leqslant r \leqslant n+1$ *and frequencies in the normalized annulus,*

$$\left| \bigwedge_{j=1}^{r} v(z;\xi_j) \right| \asymp \left| \bigwedge_{j=1}^{r} G(z;\xi_j) \right|.$$

The same statements hold for the rescaled phase ϕ^λ. *The quadratic-cone condition is preserved by the changes of variables in the local and Lorentz reductions of* [4].

Proof. Differentiating (2.4) in ξ_j gives

$$\mathbf{Q}_z v(z;\xi) \cdot \partial_{\xi_j} v(z;\xi) = 0.$$

Since $\partial_\xi v$ has rank n, its normal space is one-dimensional. This proves (2.9). The wedge-product comparison follows from the uniform bounds for $\mathbf{Q}_z$, its inverse, and $|v(z;\xi)| \asymp 1$.

The uniform bounds for the defining quadratic forms and their stability under the local and Lorentz reductions follow from Lemma 2.6. □

For a Riemannian half-wave phase, the eikonal equation gives (2.4) with

$$\mathbf{Q}_{(x,t)} = \begin{pmatrix} (g^{ij}(x)) & 0 \\ 0 & -1 \end{pmatrix}.$$

The parameter reduction to local wave parametrices is as in [1].

We now record the local geometric estimate.

Lemma 2.8. *Let* $3 \leqslant k \leqslant n+1$, *let* Q *be an* $R^{1/2}$*-cube centered at* z_Q, *and let* T_i *be an* R*-plank with frequency direction* ξ_i, $1 \leqslant i \leqslant k$. *Set*

$$\vec{v}_i = \partial_z \phi^\lambda(z_Q;\xi_i),$$

and suppose that, for a fixed constant C_0,

$$|\vec{v}_1 \wedge \cdots \wedge \vec{v}_k| \gtrsim K^{-C_0}. \tag{2.10}$$

Then

$$\left| Q \cap \bigcap_{i=1}^{k} T_i \right| \lesssim K^{O(1)} R^{\frac{n+1-k}{2}}. \tag{2.11}$$

The conclusion remains valid for the $O(K^2)$-neighborhoods of the planks, with a different $K^{O(1)}$ factor.

Proof. We use the local flattening underlying [4, Lemma 5.8], retaining the translation of each slab. Let b_i be the center of the base box $T_i^\flat$. The defining equation for the core flow is $\partial_\xi \phi^\lambda(z;\xi_i) = u$. Euler's identity therefore gives

$$\phi^\lambda(z;\xi_i) = \xi_i \cdot u.$$

Since the short direction of $T_i^\flat$ is ξ_i, points in T_i satisfy

$$|\phi^\lambda(z;\xi_i) - \xi_i \cdot b_i| \lesssim R^{2\delta}.$$

For $z \in Q$, Taylor's theorem and $\|\partial_z^2 \phi^\lambda\| \lesssim \lambda^{-1}$ give

$$\phi^\lambda(z;\xi_i) = \phi^\lambda(z_Q;\xi_i) + \vec{v}_i \cdot (z - z_Q) + O(R/\lambda).$$

Since $R \leqslant \lambda^{1-\varepsilon}$, the error is $O(1)$. Thus $Q \cap T_i$ is contained in an affine slab of width $O(R^{2\delta}+1)$ with normal $\vec{v}_i$. For the $O(K^2)$-neighborhood of T_i, the width is instead $O(R^{2\delta}+K^2+1)$.

Let $V = \operatorname{span}\{\vec{v}_1,\dots,\vec{v}_k\}$ and choose an orthonormal basis $e_{k+1},\dots,e_{n+1}$ of $V^\perp$, with this list empty when $k = n+1$. The linear map

$$L(z) = \big(\vec{v}_1 \cdot z, \dots, \vec{v}_k \cdot z, e_{k+1} \cdot z, \dots, e_{n+1} \cdot z\big)$$

has determinant of absolute value $|\vec{v}_1 \wedge \cdots \wedge \vec{v}_k|$. The image of the intersection inside Q is contained in a translated box with k side lengths $O(R^{2\delta}+K^2+1)$ and $n+1-k$ side lengths $O(R^{1/2})$. Consequently,

$$\left| Q \cap \bigcap_{i=1}^{k} T_i \right| \lesssim K^{C_0}(R^{2\delta}+K^2+1)^k R^{\frac{n+1-k}{2}} \lesssim K^{O(1)} R^{\frac{n+1-k}{2}}.$$

The same argument applies to the enlarged planks. Here $R^{O(\delta)}$ is absorbed using the parameter hierarchy. □

3. Refinements

Throughout the proof of the estimate at $p = p'(n)$, we assume $n \geqslant 4$ and (2.4). The assertions without this condition are treated separately at the end of Subsection 3.4. We set

$$k = \frac{n+4}{2}.$$

We further assume that $f \in \mathcal{S}(\mathbb{R}^n)$ satisfies (2.3) and is concentrated on a collection of wave packets $\mathbb{T}$ such that

$$\|f_T\|_2^2 \sim R^{\frac{n-1}{2}}$$

for every $T \in \mathbb{T}$. Finally, we set $p = p'(n)$.

3.1. **A Refined Two-Ends Reduction.** In this subsection, we refine the two-ends reduction introduced in [4] by separating the k-broad structure used to control the L^p norm from the 2-broad structure needed in the hairbrush argument. This is achieved by introducing a refined broad cardinality and redefining the parameter μ accordingly, which allows us to obtain a stronger incidence estimate.

Let $K_\circ = R^{\varepsilon^{100}}$. As in [4, Section 6], use nested grids at scales K^2, $R^{1/2}$, and $R/K_\circ$, adjusting these scales by bounded factors if necessary. In the combinatorial arguments the "balls" B are half-open K^2-cubes; analytical estimates are applied on comparable balls. Each B lies in a unique $R^{1/2}$-cube Q, and each Q lies in a unique horizontal region

$$\omega = \mathbb{R}^n \times I, \qquad |I| \sim R/K_\circ.$$

Let Ω denote this partition into horizontal regions. This convention avoids boundary ambiguity in all occurrences of $B \subset Q$ and $B \subset \omega$. We recall the following notions from [4, Section 6].

Definition 3.1 (Shaded incidence triple)**.** [4, Definition 6.1] *Let Ω be the collection of horizontal regions defined above. A **shaded incidence triple**, or simply a **triple**, $(\mathbf{B}, \mathbf{T}; \mathcal{G})$ consists of the following:*

(1) $\mathbf{B} = \{B\}$ is a collection of K^2-balls intersecting B_R^{n+1};
(2) $\mathbf{T} = \{T\}$ is a collection of R-planks in B_R^{n+1};
*(3) $\mathcal{G} : \mathbf{T} \to P(\Omega)$ is a **shading map**, where $P(X)$ denotes the power set of X.*

Given two shading maps $\mathcal{G}_1$ and $\mathcal{G}_2$, we write $\mathcal{G}_1 \subset \mathcal{G}_2$ if $\mathcal{G}_1(T) \subset \mathcal{G}_2(T)$ for every $T \in \mathbf{T}$.

Given a triple $(\mathbf{B}, \mathbf{T}; \mathcal{G})$, we define the incidence count

$$\mathcal{I}(\mathbf{B}, \mathbf{T}; \mathcal{G}) := \#\Big\{(B, T) \in \mathbf{B} \times \mathbf{T} : B \cap T \neq \varnothing,\ B \subset \bigcup \mathcal{G}(T)\Big\}.$$

For each $B \in \mathbf{B}$, define

$$\mathbf{T}(B; \mathcal{G}) := \Big\{T \in \mathbf{T} : B \cap T \neq \varnothing,\ B \subset \bigcup \mathcal{G}(T)\Big\},$$

and for each $T \in \mathbf{T}$, define

$$\mathbf{B}(T; \mathcal{G}) := \Big\{B \in \mathbf{B} : B \cap T \neq \varnothing,\ B \subset \bigcup \mathcal{G}(T)\Big\}.$$

Definition 3.2. [4, Definition 7.2] *Given a triple $(\mathbf{B}, \mathbf{T}; \mathcal{G})$, we define*

$$L^p(\mathbf{B}, \mathbf{T}; \mathcal{G}; f) := \sum_{B \in \mathbf{B}} \Big\| \sum_{T \in \mathbf{T}(B; \mathcal{G})} \mathcal{F}^\lambda f_T \Big\|_{L^p(B)}^p.$$

Definition 3.3 (Broad incidence)**.** [4, Definition 6.3] *Given a triple $(\mathbf{B}, \mathbf{T}; \mathcal{G})$, define*

$$\mathcal{I}_{\mathrm{br}}(\mathbf{B}, \mathbf{T}; \mathcal{G}) := \inf_{\tau \in \Theta_{K^{-1}}} \mathcal{I}\big(\mathbf{B}, \{T \in \mathbf{T} : \theta(T) \not\Subset 3\tau\}; \mathcal{G}\big).$$

For any angular set S, write $\mathbf{T}[S] = \{T \in \mathbf{T} : \theta(T) \subset S\}$ and $\mathbf{T}_\tau = \mathbf{T}[\tau]$. In the 2-broad definitions below we use the fixed enlargement 3τ, as in [4, Definition 6.3]; this supplies the angular separation needed in the hairbrush estimate. Fixed changes of the enlargement constant have no effect on the $K^{O(1)}$ losses.

We next introduce a refined version of the broad cardinality from [4]. Fix an integer $A \geqslant (C_n \log K)^2$, with C_n sufficiently large. For $j \geqslant 3$, the definition below is made at a fixed base point z; when applied to $\mathbf{T}(B;\mathcal{G})$ we take $z = z_B$ and suppress this dependence. For a collection of planks $\mathbf{T}$, we define the broad subcollection $\mathbf{T}_{\mathrm{br},j} \subset \mathbf{T}$ as follows.

If $j = 2$, choose $\tau_0 \in \Theta_{K^{-1}}$ such that

$$\#\big\{T \in \mathbf{T} : \theta(T) \not\Subset 3\tau_0\big\} = \min_{\tau \in \Theta_{K^{-1}}} \#\big\{T \in \mathbf{T} : \theta(T) \not\Subset 3\tau\big\},$$

and set

$$\mathbf{T}_{\mathrm{br},2} := \big\{T \in \mathbf{T} : \theta(T) \not\Subset 3\tau_0\big\}.$$

If $j \geqslant 3$, choose $(j-1)$-dimensional subspaces $V_1, \dots, V_A$ such that

$$\max_{\substack{\tau \in \Theta_{K^{-1}} \\ \tau \notin V_i,\ 1 \leqslant i \leqslant A}} \#\big\{T \in \mathbf{T} : \theta(T) \subset \tau\big\}$$

is minimized, and then choose $\tau_0 \in \Theta_{K^{-1}}$ satisfying $\tau_0 \notin V_i$ for all $1 \leqslant i \leqslant A$ such that

$$\#\big\{T \in \mathbf{T} : \theta(T) \subset \tau_0\big\} = \max_{\substack{\tau \in \Theta_{K^{-1}} \\ \tau \notin V_i,\ 1 \leqslant i \leqslant A}} \#\big\{T \in \mathbf{T} : \theta(T) \subset \tau\big\}.$$

We then set

$$\mathbf{T}_{\mathrm{br},j} := \big\{T \in \mathbf{T} : \theta(T) \subset \tau_0\big\}.$$

Accordingly,

$$\#_{\mathrm{br},j}\mathbf{T} := \#\mathbf{T}_{\mathrm{br},j}.$$

If the family of admissible caps is empty, set the broad cardinality to zero and take the broad subcollection to be empty. In all other cases, fix arbitrary minimizing and maximizing choices once and for all.

3.1.1. *Algorithm 1.* We now describe Algorithm 1, which provides a procedure for refining a given triple $(\mathbf{B}, \mathbf{T}; \mathcal{G})$, where $\mathbf{T}$ denotes the collection of wave packets and $\mathbf{B}$ denotes the integration domain. The output consists of a refined triple together with several auxiliary parameters.

Input: Let $(\mathbf{B}, \mathbf{T}; \mathcal{G})$ be an incidence triple, and suppose that there exists $\nu > 0$ such that, for every $B \in \mathbf{B}$,

$$\Big\| \sum_{T \in \mathbf{T}(B;\mathcal{G})} \mathcal{F}^{\lambda} f_T \Big\|_{L^p(B)} \sim \nu.$$

Pigeonholing and pruning: Pigeonhole the broad cardinality into the class zero and the positive dyadic classes. If the zero class dominates, retain it as a terminal narrow output: its local k-broad contribution vanishes. After the horizontal localization in Subsection 3.1.3, this output is treated by the narrow argument of [4, Section 8.3], without any use of negative powers of μ. The statements that follow concern the positive class. By dyadic pigeonholing, there exist $\mu \lesssim R^{O(1)}$ and a subcollection $\mathbf{B}_1 \subset \mathbf{B}$ such that

$$\#_{\mathrm{br},k}\mathbf{T}(B;\mathcal{G}) \sim \mu \qquad \text{for all } B \in \mathbf{B}_1,$$

and

$$L^p(\mathbf{B}, \mathbf{T}; \mathcal{G}; f) \lessapprox L^p(\mathbf{B}_1, \mathbf{T}; \mathcal{G}; f).$$

For dyadic numbers $W \lesssim R^{O(1)}$ and $\beta \lesssim K_\circ$, define

$$\mathcal{G}_W(T) := \{\omega \in \mathcal{G}(T) : \#\{B \in \mathbf{B}_1(T;\mathcal{G}) : B \subset \omega\} \sim W\},$$

and

$$\mathbf{T}_{W,\beta} := \{T \in \mathbf{T} : \#\mathcal{G}_W(T) \sim \beta\}.$$

Fix $B \in \mathbf{B}_1$. By dyadic pigeonholing, there exist $W(B)$ and $\beta(B)$ such that

$$\nu \sim \Big\| \sum_{T \in \mathbf{T}(B;\mathcal{G})} \mathcal{F}^\lambda f_T \Big\|_{L^p(B)} \lessapprox \Big\| \sum_{T \in \mathbf{T}_{W(B),\beta(B)}(B;\mathcal{G}_{W(B)})} \mathcal{F}^\lambda f_T \Big\|_{L^p(B)}.$$

Applying dyadic pigeonholing once more, we may find dyadic numbers W, β, and ν_1, together with a subcollection $\mathbf{B}_2 \subset \mathbf{B}_1$, such that

$$W(B) \sim W, \qquad \beta(B) \sim \beta,$$

$$\Big\| \sum_{T \in \mathbf{T}_{W,\beta}(B;\mathcal{G}_W)} \mathcal{F}^\lambda f_T \Big\|_{L^p(B)} \sim \nu_1 \quad \text{for all } B \in \mathbf{B}_2.$$

and

$$L^p(\mathbf{B}_1, \mathbf{T}; \mathcal{G}; f) \lessapprox L^p(\mathbf{B}_2, \mathbf{T}_{W,\beta}; \mathcal{G}_W; f).$$

Finally, use the fixed $R^{1/2}$-grid in B_R^{n+1}. By dyadic pigeonholing, there exist dyadic numbers m, $\bar{m}$, and σ, together with a collection $\mathcal{Q}$ of $R^{1/2}$-cubes, such that, for every $Q \in \mathcal{Q}$,

$$\#_{\mathrm{br},2} \bigcup_{\substack{B \in \mathbf{B}_2 \\ B \subset Q}} \mathbf{T}_{W,\beta}(B; \mathcal{G}_W) \sim m,$$

$$\#_{\mathrm{br},2} \bigcup_{\substack{B \in \mathbf{B}_2 \\ B \subset Q}} \mathbf{T}(B; \mathcal{G}) \sim \bar{m},$$

and

$$\#\{B \in \mathbf{B}_2, B \subset Q\} \sim \sigma.$$

Moreover,

$$L^p(\mathbf{B}_2, \mathbf{T}_{W,\beta}; \mathcal{G}_W; f) \lessapprox L^p(\mathbf{B}_3, \mathbf{T}_{W,\beta}; \mathcal{G}_W; f),$$

where

$$\mathbf{B}_3 := \{B \in \mathbf{B}_2 : B \subset Q \text{ for some } Q \in \mathcal{Q}\}$$

Finally, define

$$l := \sup_{T \in \mathbf{T}_{W,\beta}} \#\{Q \in \mathcal{Q} : Q \cap T \neq \varnothing\}.$$

We now conclude Algorithm 1 and record several properties of the quantities introduced above. By construction, we trivially have

$$\mu \lesssim \bar{m}, \qquad m \lesssim \bar{m}, \tag{3.1}$$

as well as

$$\mathcal{I}(\mathbf{B}_1, \mathbf{T}; \mathcal{G}) \gtrsim \#\mathbf{B}_1 \, \mu, \tag{3.2}$$

and

$$\mathcal{I}(\mathbf{B}_1, \mathbf{T}_{W,\beta}; \mathcal{G}_W) \lesssim \#\mathbf{T}_{W,\beta} \, \beta W. \tag{3.3}$$

Using the hairbrush argument, we also have the following estimate.

Lemma 3.4. [4, Lemma 6.5] *Suppose that $\beta > 100$. Then*

$$\#\mathbf{B}_1 \gtrsim (KK_\circ)^{-O(1)} Wlm.$$

We now apply the multilinear geometric estimate to obtain an improved upper bound for σ.

Lemma 3.5. *Under the assumptions of this section, including* (2.4)*, and with A sufficiently large,*

$$\sigma \lesssim K^{O(1)} R^{\frac{n+1-k}{2}} \bar{m}^k \mu^{-k}.$$

Proof. Fix $Q \in \mathcal{Q}$ and let z_Q be its center. Set

$$\mathcal{T}_0 = \bigcup_{\substack{B \in \mathbf{B}_3 \\ B \subset Q}} \mathbf{T}(B; \mathcal{G}), \qquad \mathcal{T} = (\mathcal{T}_0)_{\mathrm{br},2}.$$

The nested-grid convention gives

$$\#\mathcal{T} \sim \bar{m}, \qquad \#\{B \in \mathbf{B}_3 : B \subset Q\} \sim \sigma.$$

There is a cap τ_0 such that

$$\mathcal{T} = \{T \in \mathcal{T}_0 : \theta(T) \not\Subset 3\tau_0\}.$$

We show that each such B supplies k subcollections of $\mathcal{T}(B; \mathcal{G})$, each of cardinality at least $K^{-O(1)}\mu$, whose actual packet normals satisfy (2.10) uniformly.

Fix $B \subset Q$, with center z_B. The coarse caps meeting $3\tau_0$ form a family of cardinality at most C_n. For each of these caps, choose a $(k-1)$-plane containing one of its Gauss directions at z_B, and include these planes among the exceptions in the broad-cardinality definition. Recall that the angle between a cap and a plane is the infimum over all Gauss directions in that cap. Thus every cap admissible for this list is disjoint from $3\tau_0$, and its packets survive the pruning.

By $\#_{\mathrm{br},k}\mathbf{T}(B; \mathcal{G}) \sim \mu$, every choice of at most A exceptional planes leaves an admissible coarse cap containing $\gtrsim \mu$ packets. We choose k rich subcollections inductively. Suppose that $j < k$ representative frequencies $\xi_1, \dots, \xi_j$ have been selected. Choose a $(k-1)$-plane V_j containing

$$\operatorname{span}\{G^\lambda(z_B; \xi_1), \dots, G^\lambda(z_B; \xi_j)\},$$

with an arbitrary choice when $j = 0$. Add V_j to the fixed list of planes excluding the caps that meet $3\tau_0$. This uses at most $C_n + 1$ planes, so the choice is allowed when $A \geqslant C_n + 1$.

There is consequently an admissible coarse cap τ_{j+1} containing $\gtrsim \mu$ packets, all with Gauss directions at angle greater than K^{-2} from V_j. Subdivide τ_{j+1} into angular subcaps of radius K^{-M}, where M is a sufficiently large fixed constant depending only on n and the normalized phase bounds. Retain one containing $\gtrsim K^{-O(1)}\mu$ packets, denote this subcollection by $\mathcal{T}_{j+1}(B)$, and choose ξ_{j+1} to be the frequency direction of one of its packets.

After k steps, the representative Gauss directions satisfy

$$\left| \bigwedge_{j=1}^{k} G^\lambda(z_B; \xi_j) \right| \gtrsim K^{-2(k-1)}.$$

By Lemma 2.7, the same polynomial lower bound holds for their slab normals. Taking M sufficiently large makes this bound uniform over all choices of actual packet directions from $\mathcal{T}_1(B), \dots, \mathcal{T}_k(B)$.

On Q, the variation of the slab normals is $O(R^{1/2}/\lambda) \lesssim R^{-1/2}$. By choosing ε sufficiently small relative to the fixed powers of K, the transversality therefore persists at z_Q. Fix a constant C_0 for which the preceding lower bounds imply

(2.10), and let $\mathcal{A}_Q$ be the set of ordered k-tuples of packets in $\mathcal{T}$ satisfying that condition at z_Q.

Write $T^+ = N_{CK^2}(T)$, with C large enough that $B \cap T \neq \varnothing$ implies $B \subset T^+$. For every $B \in \mathbf{B}_3$ contained in Q, the preceding choices give at least $K^{-O(1)}\mu^k$ tuples in $\mathcal{A}_Q$, and each corresponding enlarged intersection contains B. The B's are disjoint grid cubes of common volume $v_K \sim K^{2(n+1)}$. Hence

$$\begin{aligned} K^{-O(1)}\mu^k \sigma v_K &\lesssim \sum_{(T_1,\dots,T_k)\in\mathcal{A}_Q} \left| Q \cap \bigcap_{i=1}^{k} T_i^+ \right| \\ &\lesssim K^{O(1)} R^{\frac{n+1-k}{2}} \#\mathcal{A}_Q \\ &\leqslant K^{O(1)} R^{\frac{n+1-k}{2}} (\#\mathcal{T})^k. \end{aligned}$$

The second inequality is Lemma 2.8. Absorbing the powers of K and using $\#\mathcal{T} \sim \bar{m}$ proves the lemma. □

3.1.2. *Algorithm 2.* Algorithm 2 repeatedly applies Algorithm 1 to obtain additional quantitative relationships among the parameters. The same iterative strategy was previously used in [11]. By dyadic pigeonholing, we may assume that there exists a finitely overlapping collection $\mathcal{B} = \{B\}$ of K^2-balls such that the quantities $\|\mathcal{F}^\lambda f\|_{L^p(B)}$ are comparable for all $B \in \mathcal{B}$, and

$$\|\mathcal{F}^\lambda f\|^p_{L^p(B_R^{n+1})} \lessapprox \sum_{B\in\mathcal{B}} \|\mathcal{F}^\lambda f\|^p_{L^p(B)}.$$

Set

$$\mathbb{T}^{(0)} = \mathbb{T}, \qquad \mathcal{B}^{(0)} = \mathcal{B},$$

and define

$$\mathcal{G}^{(0)}(T) = \Omega \qquad \text{for every } T \in \mathbb{T}.$$

We now apply Algorithm 1 iteratively, treating a terminal zero-broad output by the narrow argument described above. On the positive branch, use lower dyadic representatives for the cardinalities, so that the monotonicity statements below hold. At the i-th step, we apply Algorithm 1 to

$$(\mathcal{B}^{(i-1)}, \mathbb{T}^{(i-1)}; \mathcal{G}^{(i-1)}; f),$$

which produces the following:

(1) Parameters

$$\mu^{(i-1)},\ W^{(i-1)},\ \beta^{(i-1)},\ m^{(i-1)},\ l^{(i-1)},\ \sigma^{(i-1)}.$$

(2) Two collections of K^2-balls satisfying

$$\mathcal{B}_3^{(i-1)} \subset \mathcal{B}_1^{(i-1)} \subset \mathcal{B}^{(i-1)}.$$

(3) A collection of planks

$$\mathbb{T}^{(i-1)}_{W^{(i-1)},\beta^{(i-1)}} \subset \mathbb{T}^{(i-1)},$$

a shading map

$$\mathcal{G}^{(i-1)}_{W^{(i-1)}} \subset \mathcal{G}^{(i-1)},$$

and a collection $\mathcal{Q}^{(i-1)}$ of $R^{1/2}$-cubes.

We then set

$$\mathcal{B}^{(i)} := \mathcal{B}_3^{(i-1)}, \qquad \mathbb{T}^{(i)} := \mathbb{T}^{(i-1)}_{W^{(i-1)},\beta^{(i-1)}}, \qquad \mathcal{G}^{(i)} := \mathcal{G}^{(i-1)}_{W^{(i-1)}}.$$

By construction, for every i we have

$$\mu^{(i+1)} \leqslant \mu^{(i)} \lesssim R^{O(1)},$$
$$\bar{m}^{(i+1)} \leqslant m^{(i)} \leqslant \bar{m}^{(i)} \lesssim R^{O(1)},$$

and

$$\mathcal{I}(\mathcal{B}_1^{(i+1)}, \mathbb{T}^{(i+1)}; \mathcal{G}^{(i+1)}) \leqslant \mathcal{I}(\mathcal{B}_1^{(i)}, \mathbb{T}^{(i)}; \mathcal{G}^{(i)}) \lesssim R^{O(1)}.$$

Let $\kappa = R^{\varepsilon^{500}}$. By pigeonholing the above monotone sequences, there exists $N \lesssim_\varepsilon 1$ such that

$$\mu^{(N)} \leqslant \kappa\, \mu^{(N+1)},$$
$$\bar{m}^{(N)} \leqslant \kappa\, \bar{m}^{(N+1)} \leqslant \kappa\, m^{(N)},$$

and

$$\mathcal{I}(\mathcal{B}_1^{(N)}, \mathbb{T}^{(N)}; \mathcal{G}^{(N)}) \leqslant \kappa\, \mathcal{I}(\mathcal{B}_1^{(N+1)}, \mathbb{T}^{(N+1)}; \mathcal{G}^{(N+1)}).$$

With a slight abuse of notation, we suppress the superscripts and denote

$$\mu^{(N)},\ W^{(N)},\ \beta^{(N)},\ m^{(N)},\ l^{(N)},\ \mathcal{B}^{(N+1)},\ \mathbb{T}^{(N+1)},\ \mathcal{G}^{(N+1)},\ \sigma^{(N)},$$

and $\mathcal{Q}^{(N)}$ simply by

$$\mu,\ W,\ \beta,\ m,\ l,\ \mathcal{B},\ \mathbb{T},\ \mathcal{G},\ \sigma,$$

and $\mathcal{Q}$, respectively. We also redefine

$$f := \sum_{T \in \mathbb{T}} f_T.$$

Combining the preceding relations with (3.1), (3.2), (3.3), Lemma 3.4, and Lemma 3.5, we obtain, whenever $\beta > 100$,

$$lm\mu \lesssim K^{O(1)} \#\mathbb{T}. \tag{3.4}$$

Moreover,

$$\mu \lesssim \kappa^{O(1)} m, \tag{3.5}$$

and

$$\sigma \lesssim (K\kappa)^{O(1)} R^{\frac{n+1-k}{2}} m^k \mu^{-k}. \tag{3.6}$$

Lemma 3.6. *We have*

$$\mu \lesssim K^{O(1)} \, (\#\mathbb{T})^{1/2} \, l^{-1/2} \min\left\{1, \frac{R^{\frac{n+1-k}{2}}}{\sigma}\right\}^{\frac{1}{2k}}.$$

Proof. Combining (3.4) and (3.5), and absorbing $\kappa^{O(1)}$ into $K^{O(1)}$, we obtain

$$\mu \lesssim K^{O(1)} \, (\#\mathbb{T})^{1/2} \, l^{-1/2}. \tag{3.7}$$

On the other hand, (3.4) gives

$$m \lesssim K^{O(1)} \frac{\#\mathbb{T}}{l\mu}.$$

Substituting this into (3.6), we obtain

$$\sigma \lesssim K^{O(1)} R^{\frac{n+1-k}{2}} (\#\mathbb{T})^k l^{-k} \mu^{-2k},$$

and consequently

$$\mu \lesssim K^{O(1)} (\#\mathbb{T})^{1/2} l^{-1/2} \left(\frac{R^{\frac{n+1-k}{2}}}{\sigma} \right)^{\frac{1}{2k}}.$$

Combining this estimate with (3.7) proves the lemma. □

3.1.3. *Completing the Reduction.* Subdivide each horizontal region into grid cells B_0 of size $R/K_\circ$, compatible with the K^2-grid. We call these cells $R/K_\circ$-balls, as above. If $\beta \leqslant 100$, the proof is completed by the one-end induction of [4, Section 8.1]. A terminal zero-broad output is instead treated by the narrow induction in [4, Section 8.3]. Neither case uses Lemma 3.5. We may therefore assume that $\beta > 100$ and that the positive broad branch has been selected. The inherited reductions apply in the phase class of Theorem 2.3 by Lemma 2.7 when that additional condition is needed.

Choose a cell B_0 for which

$$\sum_{\substack{B\in\mathcal{B}\\ B\subset B_0}} \left\| \sum_{T\in\mathbb{T}(B;\mathcal{G})} \mathcal{F}^\lambda f_T \right\|^p_{L^p(B)}$$

is maximal, and let ω_0 be the unique horizontal region containing B_0. Set

$$X = \bigcup_{\substack{B\in\mathcal{B}\\ B\subset B_0}} B, \qquad \mathbb{T}_0 = \{T \in \mathbb{T} : T \cap B_0 \neq \varnothing,\ \omega_0 \in \mathcal{G}(T)\}, \qquad f_0 = \sum_{T\in\mathbb{T}_0} f_T.$$

In particular, if $B \subset B_0$, $T \in \mathbb{T}_0$, and $T \cap B \neq \varnothing$, then $T \in \mathbb{T}(B;\mathcal{G})$. This is the shaded-to-unshaded localization needed below.

Since $p = p'(n)$ satisfies (2.5), the two-ends broad–narrow reduction of [4, Section 8] leaves us to prove

$$\|\mathcal{F}^\lambda f_0\|_{BL^p_{k,3A}(X)} \lesssim R^{(n-1)(\frac{1}{2}-\frac{1}{p})+O(\varepsilon^2)} \|f\|_2^{2/p} \mathcal{W}(f, B_R^{n+1})^{1-2/p}. \tag{3.8}$$

The fixed enlargement of the broadness parameter to $3A$ is absorbed in the choice of A in that reduction.

3.2. **Applying Refined Decoupling.** Since μ has been redefined, the decoupling estimate in [4] no longer follows immediately. However, the same estimate can still be established.

We briefly explain this point. Let

$$q_n := \frac{2(n+1)}{n-1}$$

denote the decoupling exponent. For each $B \in \mathcal{B}$, by the new definition of the k-broad cardinality, there exist $(k-1)$-dimensional subspaces $V_1, \dots, V_A$ such that, for every K^{-1}-cap τ satisfying

$$\tau \notin V_j \qquad \text{for all } 1 \leqslant j \leqslant A,$$

we have

$$\#\{T \in \mathbb{T}_0, \theta(T) \subset \tau, T \cap B \neq \varnothing\} \lesssim \mu.$$

By the definition of the k-broad norm, there exists a K^{-1}-cap $\tau(B)$ satisfying

$$\tau(B) \notin V_j \qquad \text{for all } 1 \leqslant j \leqslant A,$$

such that

$$\left\|\mathcal{F}^\lambda f_0\right\|_{BL^{q_n}_{k,A}(B)} \lesssim \left\|\mathcal{F}^\lambda f_{0,\tau(B)}\right\|_{L^{q_n}(B)}.$$

Therefore, following the argument in [4], we obtain

Lemma 3.7. *We have*

$$\|\mathcal{F}^\lambda f_0\|^{q_n}_{BL^{q_n}_{k,A}(X)} \lesssim K^{O(1)}R^2 \cdot \left(R^{-\frac{n-3}{2(n-1)}}\mu^{\frac{2}{n-1}}(\#\mathbb{T})^{-\frac{1}{n-1}}\right)\|f\|_2^2 \mathcal{W}(f, B_R^{n+1})^{q_n-2}.$$

The argument in [4] invokes a local constancy property for $\mathcal{F}^\lambda$. Such a property is not available in this generality for variable-coefficient Fourier integral operators. Nevertheless, the resulting estimate remains valid, since the local constancy argument can be replaced by the following bound.

Lemma 3.8. [12, Corollary 2.11] *We have*

$$\|\mathcal{F}^\lambda f_T\|_{L^{q_n}(B^{n+1}_{2R})} \lesssim \|f_T\|_2.$$

3.3. **A Refined L^2 Estimate.** We first recall the L^2 estimate from [4, Corollary 9.3].

Lemma 3.9. *We have*

$$\|\mathcal{F}^\lambda f_0\|_{BL^2_{k,A}(X)} \lessapprox K^{O(1)} l^{1/2} \sigma^{\frac{1}{n+k+1}} \|f\|_{L^2}.$$

This bound becomes ineffective when σ is large. To obtain a refinement in this regime, we invoke [12, Corollary 2.12].

Lemma 3.10. *Let Q be an $R^{1/2}$-ball and let $\mathbf{T}$ be a collection of wave packets. Then*

$$\left\|\sum_{T\in\mathbf{T}} \mathcal{F}^\lambda f_T\right\|^2_{L^2(Q)} \lesssim R^{1/2} \sum_{T\in\mathbf{T}} \|f_T\|_2^2.$$

For each $Q \in \mathcal{Q}$, apply Lemma 3.10 to the coarse-cap pieces and sum. The definition of the broad norm gives

$$\|\mathcal{F}^\lambda f_0\|^2_{BL^2_{k,A}(Q)} \lesssim R^{1/2} \sum_{\substack{T\in\mathbb{T}_0 \\ T\cap Q\neq\varnothing}} \|f_T\|_2^2.$$

Since each $T \in \mathbb{T}_0$ intersects at most l cubes in $\mathcal{Q}$, summing over $Q \in \mathcal{Q}$ yields

$$\begin{aligned}\|\mathcal{F}^\lambda f_0\|^2_{BL^2_{k,A}(X)} &\lesssim \sum_{Q\in\mathcal{Q}} \|\mathcal{F}^\lambda f_0\|^2_{BL^2_{k,A}(Q)} \\ &\lesssim l R^{1/2} \|f\|_2^2.\end{aligned}$$

Combining this estimate with Lemma 3.9, we obtain the following refinement.

Lemma 3.11. *We have*

$$\|\mathcal{F}^\lambda f_0\|_{BL^2_{k,A}(X)} \lessapprox K^{O(1)} l^{1/2} \min\left\{\sigma^{\frac{1}{n+k+1}}, R^{1/4}\right\} \|f\|_{L^2}.$$

3.4. **Interpolation.** Using Lemma 3.6, we interpolate Lemmas 3.7 and 3.11 with the following estimate from [4].

Lemma 3.12. [4, Corollary 9.2] *Let*

$$r_n = \frac{2(n+k+1)}{n+k-1}.$$

Then

$$\|\mathcal{F}^\lambda f_0\|_{BL^{r_n}_{k,A}(X)} \lessapprox R^{n\left(\frac{1}{2}-\frac{1}{r_n}\right)} \|f\|_2^{\frac{2}{r_n}} \mathcal{W}(f, B_R^{n+1})^{1-\frac{2}{r_n}}.$$

We first prove the assertion under (2.4). Recall that

$$k = \frac{n+4}{2}, \qquad q_n = \frac{2(n+1)}{n-1}, \qquad r_n = \frac{6(n+2)}{3n+2}.$$

Set $N := \#\mathbb{T}$. Lemma 3.6 gives

$$\mu \lesssim K^{O(1)} N^{1/2} l^{-1/2} \min\left\{1, \frac{R^{(n-2)/4}}{\sigma}\right\}^{\frac{1}{n+4}}.$$

Taking the q_n-th root in Lemma 3.7 and applying this bound, we obtain

$$\begin{aligned}\|\mathcal{F}^\lambda f_0\|_{BL^{q_n}_{k,A}(X)} \lesssim\ & K^{O(1)} R^{(n-1)\left(\frac{1}{2}-\frac{1}{q_n}\right)} R^{-\frac{n-3}{4(n+1)}} l^{-\frac{1}{2(n+1)}} \\ & \times \min\left\{1, \frac{R^{(n-2)/4}}{\sigma}\right\}^{\frac{1}{(n+1)(n+4)}} \|f\|_2^{\frac{2}{q_n}} \mathcal{W}(f, B_R^{n+1})^{1-\frac{2}{q_n}}.\end{aligned}$$

Lemma 3.12 can be written as

$$\begin{aligned}\|\mathcal{F}^\lambda f_0\|_{BL^{r_n}_{k,A}(X)} \lessapprox\ & R^{(n-1)\left(\frac{1}{2}-\frac{1}{r_n}\right)} R^{\frac{2}{3(n+2)}} \\ & \times \|f\|_2^{\frac{2}{r_n}} \mathcal{W}(f, B_R^{n+1})^{1-\frac{2}{r_n}},\end{aligned}$$

while Lemma 3.11 gives

$$\|\mathcal{F}^\lambda f_0\|_{BL^2_{k,A}(X)} \lessapprox K^{O(1)} l^{1/2} \min\left\{\sigma^{\frac{2}{3(n+2)}}, R^{1/4}\right\} \|f\|_2.$$

Choose

$$\theta_q = \frac{16(n+1)(n+4)}{6n^3+25n^2+44n+4}, \qquad \theta_2 = \frac{24(n+2)}{6n^3+25n^2+44n+4},$$

$$\theta_r = \frac{3(n+2)(2n^2-n-18)}{6n^3+25n^2+44n+4}.$$

These coefficients are nonnegative for every $n \geqslant 4$ and satisfy

$$\theta_q + \theta_r + \theta_2 = 1, \qquad \frac{1}{p'(n)} = \frac{\theta_q}{q_n} + \frac{\theta_r}{r_n} + \frac{\theta_2}{2}.$$

Writing $p = p'(n)$, interpolation therefore yields

$$\|\mathcal{F}^\lambda f_0\|_{BL^p_{k,3A}(X)} \lessapprox K^{O(1)} R^{(n-1)\left(\frac{1}{2}-\frac{1}{p}\right)} E \|f\|_2^{\frac{2}{p}} \mathcal{W}(f, B_R^{n+1})^{1-\frac{2}{p}},$$

where

$$\begin{aligned}E = & R^{-\frac{n-3}{4(n+1)}\theta_q + \frac{2}{3(n+2)}\theta_r} l^{\frac{1}{2}\left(\theta_2 - \frac{\theta_q}{n+1}\right)} \\ & \times \min\left\{1, \frac{R^{(n-2)/4}}{\sigma}\right\}^{\frac{\theta_q}{(n+1)(n+4)}} \min\left\{\sigma^{\frac{2}{3(n+2)}}, R^{1/4}\right\}^{\theta_2}.\end{aligned} \tag{3.9}$$

Since

$$\theta_2 - \frac{\theta_q}{n+1} = \frac{8(n-2)}{6n^3+25n^2+44n+4} > 0,$$

the bound $l \lesssim R^{1/2}$ controls the l-factor in (3.9) by

$$R^{\frac{1}{4}\left(\theta_2 - \frac{\theta_q}{n+1}\right)}.$$

Moreover,

$$\frac{\theta_q}{(n+1)(n+4)} = \frac{2\theta_2}{3(n+2)} = \frac{16}{6n^3+25n^2+44n+4}.$$

Thus the product of the two σ-dependent factors is at most

$$\begin{aligned}\min\left\{1, \frac{R^{(n-2)/4}}{\sigma}\right\}^{\frac{2\theta_2}{3(n+2)}} \sigma^{\frac{2\theta_2}{3(n+2)}} \\ = \min\{\sigma, R^{(n-2)/4}\}^{\frac{2\theta_2}{3(n+2)}} \leqslant R^{\frac{n-2}{6(n+2)}\theta_2}.\end{aligned}$$

For the chosen weights,

$$-\frac{n-3}{4(n+1)}\theta_q + \frac{2}{3(n+2)}\theta_r + \frac{1}{4}\left(\theta_2 - \frac{\theta_q}{n+1}\right) + \frac{n-2}{6(n+2)}\theta_2 = 0.$$

Consequently, $E \lesssim 1$, and

$$\|\mathcal{F}^\lambda f_0\|_{BL^p_{k,3A}(X)} \lesssim R^{(n-1)\left(\frac{1}{2}-\frac{1}{p}\right)+O(\varepsilon^2)} \|f\|_2^{\frac{2}{p}} \mathcal{W}(f, B_R^{n+1})^{1-\frac{2}{p}}.$$

Here the factors $K^{O(1)}$, $\kappa^{O(1)}$, and the arbitrarily small losses have been absorbed into $R^{O(\varepsilon^2)}$. This proves (3.8) under (2.4). In particular, when $n = 4$, the weights are

$$(\theta_q, \theta_r, \theta_2) = \left(\frac{160}{241}, \frac{45}{241}, \frac{36}{241}\right),$$

and $p = p'(4) = 482/167$.

These estimates extend to larger p by interpolation with

$$\|\mathcal{F}^\lambda f\|_{L^\infty(B_R^{n+1})} \lessapprox R^{(n-1)/2} \mathcal{W}(f, B_R^{n+1}).$$

The latter follows from the finest-scale test in the density definition, the single-packet Cauchy–Schwarz bound, and the bounded spatial overlap of packets of a fixed direction. This completes the proof of Theorem 2.3.

4. Limitations

In this section, we prove Theorems 2.4 and 2.5 by constructing a nearly extremal configuration of wave packets in the Euclidean setting. The same configuration reveals two related geometric obstructions: it nearly saturates the original wave packet density estimate, while also showing that the anisotropic multiscale refinement does not completely eliminate the underlying obstruction. Thus, even in flat space, wave packet density methods encounter genuine limitations in the local smoothing problem.

The construction arose from an attempt to combine the polynomial Wolff axioms with wave packet density estimates in order to improve Euclidean local smoothing bounds. During this investigation, we encountered degenerate configurations that appeared particularly resistant to further improvement, notably when a large family of wave packets is tangent to a hyperbolic-type hypersurface. While testing

possible refinements with the assistance of ChatGPT, we observed that such configurations are close to saturating the relevant density bounds. This led to the nearly extremal example developed below and motivated our analysis of both the original and multiscale wave packet density estimates.

4.1. **Proof of Theorems 2.4.** We treat the odd- and even-dimensional cases simultaneously. Define

$$(l,\beta)=\begin{cases}\left(\dfrac{n+3}{2},\dfrac{1}{2}\right), & n \text{ odd},\\ \left(\dfrac{n}{2}+2,0\right), & n \text{ even}.\end{cases}$$

Thus $3\leqslant l\leqslant n$.

Fix a sufficiently small constant $c>0$, and consider the l-dimensional hyperboloid

$$\mathcal{H}_R=\left\{(x',x'',t)\in\mathbb{R}^l\times\mathbb{R}^{n-l}\times\mathbb{R}:|x'|^2-t^2=c^2R^2,\quad x''=0\right\}.$$

Step 1. Construction of the wave packets.

Let $\Omega_R\subset S^{l-1}$ be a maximal $CR^{-1/2}$-separated set, where C is a sufficiently large fixed constant. Then

$$\#\Omega_R\sim R^{(l-1)/2}. \tag{4.1}$$

For each $\omega\in\Omega_R$, let

$$U_\omega\subset S^{l-1}\cap\omega^\perp\cong S^{l-2}$$

be a maximal $CR^{-1/2}$-separated set. Hence

$$\#U_\omega\sim R^{(l-2)/2}. \tag{4.2}$$

Finally, let Q_R be a C-separated subset of $[-c_1R^\beta,c_1R^\beta]$ of maximal cardinality, where $c_1>0$ is sufficiently small. Thus

$$\#Q_R\sim R^\beta. \tag{4.3}$$

When n is even, $\beta=0$, and we may simply take $Q_R=\{0\}$.

For

$$\omega\in\Omega_R,\qquad u\in U_\omega,\qquad q\in Q_R,$$

define

$$b_{\omega,u,q}:=cR\,u+q\omega\in\mathbb{R}^l\subset\mathbb{R}^n.$$

Since $u\perp\omega$,

$$b_{\omega,u,q}\cdot\omega=q,\qquad |b_{\omega,u,q}|^2=c^2R^2+q^2. \tag{4.4}$$

Let θ_ω be an $R^{-1/2}$-cap centered at the direction ω. For every triple (ω,u,q), choose a standard R-scale wave packet $f_{\omega,u,q}$ associated with the phase-space tile $(\theta_\omega,b_{\omega,u,q})$, and normalize it so that

$$\|f_{\omega,u,q}\|_2\sim 1.$$

At time $t=0$, its physical footprint has dimensions $1\times R^{1/2}\times\cdots\times R^{1/2}$ with its short direction parallel to ω. Its spacetime extension is essentially supported on a plank of dimensions $1\times R^{1/2}\times\cdots\times R^{1/2}\times R$ whose center line is

$$L_{\omega,u,q}=\{(b_{\omega,u,q}-t\omega,0,t):|t|\lesssim R\}.$$

We claim that these wave packets are tangent to the $R^{1/2}$-neighborhood of $\mathcal{H}_R$.

Indeed, along the center line,

$$|b_{\omega,u,q} - t\omega|^2 - t^2 - c^2R^2 = |b_{\omega,u,q}|^2 - c^2R^2 - 2t\, b_{\omega,u,q}\cdot\omega$$
$$= q^2 - 2tq. \tag{4.5}$$

Since

$$|q| \lesssim R^\beta \leqslant R^{1/2}, \qquad |t| \lesssim R,$$

we have

$$|q^2 - 2tq| \lesssim R^{3/2}.$$

On the relevant fixed portion of the hyperboloid,

$$|\nabla(|x'|^2 - t^2 - c^2R^2)| \sim R.$$

It follows that

$$\operatorname{dist}\big(L_{\omega,u,q}, \mathcal{H}_R\big) \lesssim R^{1/2}.$$

Moreover, the direction of the center line is $(-\omega, 0, 1)$. If $z = (x', 0, t)$ is a nearby point on $\mathcal{H}_R$, then a normal vector is $N_z = (x', 0, -t)$. Using (4.4) and (4.5), one obtains

$$\angle\big((-\omega,0,1), T_z\mathcal{H}_R\big) \lesssim \frac{|q|}{R} + R^{-1/2} \lesssim R^{-1/2}.$$

Thus the usual $R^{1/2+O(\delta)}$-thickened wave packets satisfy the standard tangency condition.

Notice that when n is even, $q = 0$, and the center lines are in fact exact rulings of the hyperboloid.

Define

$$f_R = \sum_{\omega\in\Omega_R}\sum_{u\in U_\omega}\sum_{q\in Q_R} \varepsilon_{\omega,u,q} f_{\omega,u,q},$$

where $\varepsilon_{\omega,u,q} \in \{-1, 1\}$ will be chosen later.

By (4.1), (4.2), and (4.3),

$$\|f_R\|_2^2 \sim \#\mathbb{T}_R \sim R^{l-\frac{3}{2}+\beta} = R^{\frac{n+1}{2}}. \tag{4.6}$$

Step 2. The wave packet density.

Let $R^{-1/2} \leqslant s \leqslant 1$, let τ be an s-cap, and let

$$V \parallel V_{\tau,R}, \qquad V_{\tau,R} \sim Rs^2 \times (Rs)^{n-1}.$$

Thus

$$|V| \sim R^n s^{n+1}. \tag{4.7}$$

We estimate the number of selected finest-scale packets satisfying $\theta_\omega \subset \tau$ and whose initial footprints lie in V.

Let ω_τ denote the center direction of τ, and let b_V denote the center of V. Suppose that $\theta_\omega \subset \tau$ and that, for some $u \in U_\omega$ and $q \in Q_R$, the corresponding initial footprint is contained in V. Then we may write

$$b_{\omega,u,q} - b_V = a\omega_\tau + y, \qquad |a| \lesssim Rs^2, \quad |y| \lesssim Rs,$$

where $y \perp \omega_\tau$. Since $\angle(\omega, \omega_\tau) \lesssim s$, we obtain

$$|(b_{\omega,u,q} - b_V)\cdot\omega| \lesssim Rs^2.$$

Using (4.4), it follows that

$$|b_V\cdot\omega| \lesssim R^\beta + Rs^2 \lesssim R^{1/2} + Rs^2.$$

Let $E = \mathbb{R}^l \times \{0\} \subset \mathbb{R}^n$. For s smaller than a sufficiently small fixed constant, the compatibility condition and (4.4) imply $|P_E b_V| \sim R$. Since $\omega \in S^{l-1} \subset E$, the preceding estimate shows that the admissible directions ω lie in an $O(R^{-1/2} + s^2)$-neighborhood of a codimension-one great sphere inside the s-cap. Since the points of Ω_R are $CR^{-1/2}$-separated, we obtain

$$\#\left\{\omega \in \Omega_R : \begin{array}{l} \theta_\omega \subset \tau, \\ \omega \text{ is compatible with } V \end{array}\right\} \lesssim R^{\frac{l-1}{2}} s^{l-2} \left(R^{-1/2} + s^2\right). \tag{4.8}$$

For s bounded below by a sufficiently small fixed constant, the same estimate follows from the trivial bound

$$\#\{\omega \in \Omega_R : \theta_\omega \subset \tau\} \lesssim (R^{1/2}s)^{l-1}.$$

Fix such an ω. The points $\{cR\,u : u \in U_\omega\}$ are $R^{1/2}$-separated on a sphere of radius comparable to R. If the corresponding footprints are contained in the same box V, their u-parameters must lie in an $O(s)$-cap on S^{l-2}. Hence

$$\#\{u \in U_\omega : \text{the corresponding center is compatible with } V\} \lesssim (R^{1/2}s)^{l-2}. \tag{4.9}$$

Finally, changing q moves the center essentially in the short direction of V. Since the short side of V has length Rs^2,

$$\#\{q \in Q_R : b_{\omega,u,q} \text{ is compatible with } V\} \lesssim \min\{R^\beta, Rs^2\}. \tag{4.10}$$

Combining (4.8)–(4.10), we obtain

$$\#\left\{T \in \mathbb{T}_R : \theta(T) \subset \tau,\ T^\flat \subset V\right\} \lesssim R^{l-\frac{3}{2}} s^{2l-4} \left(R^{-1/2} + s^2\right) \min\{R^\beta, Rs^2\}.$$

Since every selected packet has L^2-mass comparable to 1, the definition of wave packet density and (4.7) yield

$$\mathcal{W}(f_R, B_R^{n+1})^2 \lesssim \sup_{R^{-1/2} \leqslant s \leqslant 1} R^{l-n-\frac{3}{2}} s^{2l-n-5} \left(R^{-1/2} + s^2\right) \min\{R^\beta, Rs^2\}. \tag{4.11}$$

We now evaluate this supremum separately in the two parity cases.

Odd n. Here

$$l = \frac{n+3}{2}, \qquad \beta = \frac{1}{2}.$$

Thus (4.11) gives

$$\mathcal{W}(f_R, B_R^{n+1})^2 \lesssim \sup_{R^{-1/2} \leqslant s \leqslant 1} R^{-\frac{n}{2}} s^{-2} \left(R^{-1/2} + s^2\right) \min\{R^{1/2}, Rs^2\}.$$

If $R^{-1/2} \leqslant s \leqslant R^{-1/4}$, then

$$R^{-1/2} + s^2 \lesssim R^{-1/2}, \qquad \min\{R^{1/2}, Rs^2\} = Rs^2,$$

and hence

$$R^{-\frac{n}{2}} s^{-2} \left(R^{-1/2} + s^2\right) \min\{R^{1/2}, Rs^2\} \lesssim R^{-\frac{n-1}{2}}.$$

If $R^{-1/4} \leqslant s \leqslant 1$, then

$$R^{-1/2} + s^2 \lesssim s^2, \qquad \min\{R^{1/2}, Rs^2\} = R^{1/2},$$

and again

$$R^{-\frac{n}{2}} s^{-2} \left(R^{-1/2} + s^2\right) \min\{R^{1/2}, Rs^2\} \lesssim R^{-\frac{n-1}{2}}.$$

Therefore,

$$\mathcal{W}(f_R, B_R^{n+1})^2 \lesssim R^{-\frac{n-1}{2}}.$$

Even n. Here

$$l = \frac{n}{2} + 2, \qquad \beta = 0.$$

Since $Rs^2 \geqslant 1$ for $s \geqslant R^{-1/2}$, we have $\min\{1, Rs^2\} = 1$. Therefore, (4.11) gives

$$\mathcal{W}(f_R, B_R^{n+1})^2 \lesssim R^{-\frac{n-1}{2}} \sup_{R^{-1/2} \leqslant s \leqslant 1} s^{-1} \left(R^{-1/2} + s^2 \right) \lesssim R^{-\frac{n-1}{2}}.$$

Thus, in both parity cases,

$$\mathcal{W}(f_R, B_R^{n+1})^2 \lesssim R^{-\frac{n-1}{2}}. \tag{4.12}$$

Step 3. A lower bound for the linear L^p norm.

Let

$$U_R = N_{CR^{1/2}}(\mathcal{H}_R) \cap B_R^{n+1},$$

restricted, if necessary, to a fixed smooth portion of the hyperboloid on which all geometric constants are uniform.

Since $\mathcal{H}_R$ is l-dimensional at spatial scale R,

$$|U_R| \sim R^l (R^{1/2})^{n+1-l} = R^{\frac{n+l+1}{2}}. \tag{4.13}$$

Each selected wave packet spends a time interval of length comparable to R inside U_R. Consequently,

$$\|e^{it\sqrt{-\Delta}} f_{\omega,u,q}\|_{L^2(U_R)}^2 \gtrsim R.$$

Choose the signs $\varepsilon_{\omega,u,q}$ independently and uniformly in $\{-1, 1\}$. By orthogonality in expectation,

$$\mathbb{E}_\varepsilon \|e^{it\sqrt{-\Delta}} f_R\|_{L^2(U_R)}^2 = \sum_{\omega,u,q} \|e^{it\sqrt{-\Delta}} f_{\omega,u,q}\|_{L^2(U_R)}^2 \gtrsim R \, \#\mathbb{T}_R.$$

Hence there exists a deterministic choice of signs for which

$$\|e^{it\sqrt{-\Delta}} f_R\|_{L^2(U_R)} \gtrsim R^{\frac{1}{2}} R^{\frac{1}{2}(l - \frac{3}{2} + \beta)} = R^{\frac{l}{2} - \frac{1}{4} + \frac{\beta}{2}}.$$

For every $p \geqslant 2$, Hölder's inequality and (4.13) imply

$$\begin{aligned} \|e^{it\sqrt{-\Delta}} f_R\|_{L^p(B_R^{n+1})} &\geqslant \|e^{it\sqrt{-\Delta}} f_R\|_{L^p(U_R)} \\ &\geqslant |U_R|^{\frac{1}{p} - \frac{1}{2}} \|e^{it\sqrt{-\Delta}} f_R\|_{L^2(U_R)} \\ &\gtrsim R^{\frac{n+l+1}{2p} + \frac{l-n-2}{4} + \frac{\beta}{2}}. \end{aligned} \tag{4.14}$$

For odd n,

$$l = \frac{n+3}{2}, \qquad \beta = \frac{1}{2},$$

and therefore

$$\|e^{it\sqrt{-\Delta}} f_R\|_{L^p(B_R^{n+1})} \gtrsim R^{\frac{3n+5}{4p} - \frac{n-1}{8}}. \tag{4.15}$$

For even n,

$$l = \frac{n}{2} + 2, \qquad \beta = 0,$$

and therefore

$$\|e^{it\sqrt{-\Delta}} f_R\|_{L^p(B_R^{n+1})} \gtrsim R^{\frac{3n+6}{4p} - \frac{n}{8}}. \tag{4.16}$$

Step 4. Consequence for the mixed wave packet density estimate.

Assume that

$$\left\|e^{it\sqrt{-\Delta}}f_R\right\|_{L^p(B_R^{n+1})} \lesssim_\varepsilon R^{(n-1)\left(\frac{1}{2}-\frac{1}{p}\right)+\varepsilon}\|f_R\|_2^{\frac{2}{p}}\mathcal{W}(f_R,B_R^{n+1})^{1-\frac{2}{p}}.$$

If n is odd, then (4.12) and (4.6) give

$$\left\|e^{it\sqrt{-\Delta}}f_R\right\|_{L^p(B_R^{n+1})} \lesssim_\varepsilon R^{\varepsilon+\frac{n-1}{4}+\frac{1}{p}}.$$

Combining this with (4.15), we obtain

$$\frac{3n+5}{4p}-\frac{n-1}{8}\leqslant\frac{n-1}{4}+\frac{1}{p}.$$

Equivalently,

$$p\geqslant 2+\frac{8}{3n-3}.$$

If n is even, then (4.12) and (4.6) give

$$\left\|e^{it\sqrt{-\Delta}}f_R\right\|_{L^p(B_R^{n+1})} \lesssim_\varepsilon R^{\varepsilon+\frac{n-1}{4}+\frac{1}{p}}.$$

Combining this with (4.16), we obtain

$$\frac{3n+6}{4p}-\frac{n}{8}\leqslant\frac{n-1}{4}+\frac{1}{p}.$$

This yields

$$p\geqslant 2+\frac{8}{3n-2}.$$

4.2. **Proof of Theorems 2.5.** We use the construction in Theorem 2.4 with $n=3$. Recall that the wave packets have initial centers

$$b_{\omega,u,q}=cRu+q\omega,\qquad \omega\in\Omega_R,\quad u\in U_\omega,\quad q\in Q_R,$$

where $\Omega_R\subset S^2$ and $U_\omega\subset S^2\cap\omega^\perp$ are $CR^{-1/2}$-separated, while Q_R is a C-separated subset of $[-c_1R^{1/2},c_1R^{1/2}]$. Each packet is normalized so that $\|f_{\omega,u,q}\|_2\sim 1$. The preceding construction gives

$$\|f_R\|_2\sim R,\qquad \|e^{it\sqrt{-\Delta}}f_R\|_{L^p(B_R^4)}\gtrsim R^{\frac{7}{2p}-\frac{1}{4}},\qquad p\geqslant 2.$$

It remains to establish

$$\widetilde{\mathcal{W}}(f_R,B_R^4)^2\lesssim R^{-3/4}. \tag{4.17}$$

Fix dyadic scales $R^{-1/2}\leqslant s_2\leqslant s_1\leqslant 1$, an (s_1,s_2)-cap τ, and a corresponding test box $V\parallel V_{\tau,R}$. Thus,

$$|V|\sim(Rs_1^2)(Rs_1)(Rs_2)=R^3s_1^3s_2.$$

Write

$$N(\tau,V):=\#\big\{T\in\mathbb{T}_R:\theta(T)\subset\tau,\ T^\flat\subset V\big\}.$$

Since the directions in Ω_R are $CR^{-1/2}$-separated, the number of directions contributing to $N(\tau,V)$ is at most

$$\#\{\omega\in\Omega_R:\theta_\omega\subset\tau\}\lesssim Rs_1s_2. \tag{4.18}$$

Fix one such ω. Since $P_{\omega^\perp} b_{\omega,u,q} = cRu$, the condition $b_{\omega,u,q} \in V$ implies $cRu \in P_{\omega^\perp} V$. The latter set has diameter $O(Rs_1)$. As the points cRu are $CR^{1/2}$-separated on a circle of radius cR, the number of possible u is therefore at most

$$O(R^{1/2} s_1). \tag{4.19}$$

For each fixed pair (ω, u), there are trivially $O(R^{1/2})$ possible values of q. If $s_1 \leqslant R^{-1/4}$, let e_τ denote the unit vector along the Rs_1^2-edge of V. The alignment of the cap and its test box gives $|\omega \cdot e_\tau| \gtrsim 1$. Projecting $b_{\omega,u,q} = cRu + q\omega$ onto e_τ, the condition $b_{\omega,u,q} \in V$ confines q to an interval of length $O(Rs_1^2)$. Since the q-parameters are C-separated and $Rs_1^2 \geqslant 1$, we obtain, for all admissible s_1,

$$\#\{q \in Q_R : b_{\omega,u,q} \in V\} \lesssim \min\{R^{1/2}, Rs_1^2\}. \tag{4.20}$$

Combining (4.18)–(4.20) yields

$$N(\tau, V) \lesssim R^{3/2} s_1^2 s_2 \min\{R^{1/2}, Rs_1^2\}.$$

Since each selected packet has squared L^2 norm comparable to 1, it follows that

$$\begin{aligned} \widetilde{\mathcal{W}}(f_R, B_R^4)^2 &\lesssim \sup_{R^{-1/2} \leqslant s_2 \leqslant s_1 \leqslant 1} \frac{R^{3/2} s_1^2 s_2 \min\{R^{1/2}, Rs_1^2\}}{R^3 s_1^3 s_2} \\ &= \sup_{R^{-1/2} \leqslant s_1 \leqslant 1} R^{-3/2} s_1^{-1} \min\{R^{1/2}, Rs_1^2\}. \end{aligned}$$

The expression in the last supremum equals

$$\begin{cases} R^{-1/2} s_1, & R^{-1/2} \leqslant s_1 \leqslant R^{-1/4}, \\ R^{-1} s_1^{-1}, & R^{-1/4} \leqslant s_1 \leqslant 1. \end{cases}$$

Both expressions are bounded by $R^{-3/4}$, proving (4.17).

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